

Simple commands

```
jella_pragathi@LAPTOP-RQQGT555:~$ pwd
/home/jella_pragathi
jella_pragathi@LAPTOP-RQQGT555:~$ ls
OSSP
jella_pragathi@LAPTOP-RQQGT555:~$ date
Fri Aug  7 08:15:01 UTC 2026
jella_pragathi@LAPTOP-RQQGT555:~$ whoami
jella_pragathi
jella_pragathi@LAPTOP-RQQGT555:~$ hostname
LAPTOP-RQQGT555
jella_pragathi@LAPTOP-RQQGT555:~$ uname
Linux
jella_pragathi@LAPTOP-RQQGT555:~$ uname -r
6.18.33.2-microsoft-standard-WSL2
jella_pragathi@LAPTOP-RQQGT555:~$ id
uid=1000(jella_pragathi) gid=1000(jella_pragathi) groups=1000(jella_pragathi),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users)
jella_pragathi@LAPTOP-RQQGT555:~$ tty
/dev/pts/0jella_pragathi@LAPTOP-RQQGT555:~$ uptime
 08:15:57 up 2 min,  1 user,  load average: 0.02, 0.03, 0.00
jella_pragathi@LAPTOP-RQQGT555:~$ users
jella_pragathi@LAPTOP-RQQGT555:~$ who
jella_pragathi@LAPTOP-RQQGT555:~$ w
 08:16:23 up 3 min,  1 user,  load average: 0.01, 0.02, 0.00
USER      TTY      FROM            LOGIN@   IDLE   JCPU   PCPU   WHAT
jella_pr pts/1    -              08:13   3:02   0.04s  0.04s  -bash
jella_pragathi@LAPTOP-RQQGT555:~$ groups
jella_pragathi adm cdrom sudo dip plugdev users
```

```
jella_pragathi@LAPTOP-RQQGT555:~$ env
SHELL=/bin/bash
WSL2_GUI_APPS_ENABLED=1
WSL_DISTRO_NAME=Ubuntu
NAME=LAPTOP-RQQGT555
PWD=/home/jella_pragathi
LOGNAME=jella_pragathi
HOME=/home/jella_pragathi
LANG=C.UTF-8
WSL_INTEROP=/run/WSL/327_interop
LS_COLORS=0:di=01:34;ln=01:36;mh=00;pi=00;33;so=01:35;do=01:35;bd=00;33;01:cd=00;33;01:or=00;31;01:mi=00;su=37;41;sg=30;43;ca=00;tw=30;42;ow=34;42;st=37;
40;ex=01:32;*tar=01:31;*tgz=01:31;*arc=01:31;*arj=01:31;*taz=01:31;*lha=01:31;*lzh=01:31;*lzm=01:31;*tlz=01:31;*txz=01:31;*tzo=01:31
;*7z=01:31;*zip=01:31;*z=01:31;*dz=01:31;*gz=01:31;*lrz=01:31;*lz=01:31;*lzo=01:31;*xz=01:31;*zst=01:31;*tztst=01:31;*bz2=01:31;*bz=01:31;*tbz
=01:31;*tbz2=01:31;*tz=01:31;*deb=01:31;*rpm=01:31;*jar=01:31;*war=01:31;*ear=01:31;*sar=01:31;*rar=01:31;*alz=01:31;*zoo=01:31;*cpi
=01:31;*7z=01:31;*rz=01:31;*cab=01:31;*wim=01:31;*swm=01:31;*dwm=01:31;*esd=01:31;*avif=01:35;*jpg=01:35;*jpeg=01:35;*mjpg=01:35;*mjpeg=01:35;*
.gif=01:35;*bmp=01:35;*pbm=01:35;*pgm=01:35;*ppm=01:35;*tga=01:35;*xbm=01:35;*xpm=01:35;*tif=01:35;*tiff=01:35;*png=01:35;*svg=01:35;*svgz=01:35
;*mng=01:35;*pcx=01:35;*mov=01:35;*mpg=01:35;*mpeg=01:35;*m2v=01:35;*mkv=01:35;*webm=01:35;*webp=01:35;*ogm=01:35;*mp4=01:35;*m4v=01:35;*mp4v=0
1:35;*vob=01:35;*qt=01:35;*nuv=01:35;*wmv=01:35;*asf=01:35;*rm=01:35;*rmvb=01:35;*flc=01:35;*avi=01:35;*fli=01:35;*flv=01:35;*gl=01:35;*dl=01:3
5;*xcf=01:35;*xwd=01:35;*yuv=01:35;*cgm=01:35;*emf=01:35;*ogv=01:35;*ogx=01:35;*aac=00;36;*au=00;36;*flac=00;36;*m4a=00;36;*mid=00;36;*midi=00;
36;*mka=00;36;*mp3=00;36;*mpc=00;36;*ogg=00;36;*ra=00;36;*wav=00;36;*oga=00;36;*opus=00;36;*spx=00;36;*xspf=00;36;*~=00;90;#*=00;90;*bak=00;90;*
old=00;90;*orig=00;90;*part=00;90;*rej=00;90;*swp=00;90;*tmp=00;90;*dpgk-dist=00;90;*dpgk-old=00;90;*ucf-dist=00;90;*ucf-new=00;90;*ucf-old=00;90;
*rpmnew=00;90;*rpmorig=00;90;*rpmsave=00;90;
WAYLAND_DISPLAY=wayland-0
LESSCLOSE=/usr/bin/lesspipe %s %s
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=jella_pragathi
DISPLAY=:0
SHLVL=1
XDG_RUNTIME_DIR=/run/user/1000
WSLENV=
XDG_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/snapd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wnl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCor
poration/WindowsSubsystemForLinux_2.7.11.0_x64_8wekyb3d8bbwe:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Fil
es/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/windows/system32:/mnt/c/windows/mnt/c/windows/System32/Wbem;/m
nt/c/windows/System32/WindowsPowerShell/v1.0:/mnt/c/windows/System32/OpenSSH:/mnt/c/Users/jella_pragathi/Downloads/gcc-15.1.0/bin:/mnt/c/Program Files/Git
/cmd:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/jella_pragathi/AppData/Local/Programs/Python/Python314/Script
s:/mnt/c/Users/jella_pragathi/AppData/Local/Programs/Python/Python314:/mnt/c/Users/jella_pragathi/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/jella_p
ragathi/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/jella_pragathi/AppData/Roaming/npm:/snap/bin
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
HOSTTYPE=x86_64
```

```
jella_pragathi@LAPTOP-RQQT555:~$ printenv
SHELL=/bin/bash
WSL2_GUI_APPS_ENABLED=1
WSL_DISTRO_NAME=Ubuntu
NAME=LAPTOP-RQQT555
PWD=/home/jella_pragathi
LOGNAME=jella_pragathi
HOME=/home/jella_pragathi
LANG=C.UTF-8
WSL_INTEROP=/run/WSL/327_interop
LS_COLORS=rs=0:di=01;34:ln=01;36:wh=00;pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=00:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tztz=01;31:*.bz2=01;31:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.avif=01;35:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.webp=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mpdv=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:*.old=00;90:*.orig=00;90:*.part=00;90:*.rej=00;90:*.swp=00;90:*.tmp=00;90:*.dpkg-dist=00;90:*.dpkg-old=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:*.rpmnew=00;90:*.rpmorig=00;90:*.rpmsave=00;90:
WAYLAND_DISPLAY=wayland-0
LESSCLOSE=/usr/bin/lesspipe %s %s
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=jella_pragathi
DISPLAY=:0
SHLVL=1
XDG_RUNTIME_DIR=/run/user/1000
WLENV=
XDG_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/napd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/mnt/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForLinux.2.7.11.0_x64_8wekyb3d8bbwe:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/windows/system32:/mnt/c/windows:/mnt/c/windows/System32/Wbem:/mnt/c/windows/System32/WindowsPowerShell/v1.0:/mnt/c/windows/System32/OpensSH:/mnt/c/Users/jella_pragathi/Downloads/gcc-15.1.0/bin:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/jella_pragathi/AppData/Local/Programs/Python/Python314/Script s:/mnt/c/Users/jella_pragathi/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/jella_pragathi/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/jella_pragathi/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/jella_pragathi/AppData/Roaming/npm:/snap/bin
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
HOSTTYPE=x86_64
```

```
jella_pragathi@LAPTOP-RQQT555:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            3.8G  0  3.8G  0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none            3.8G  4.0K  3.8G  1% /mnt/wsl
drivers         129G  11G   19G  87% /usr/lib/wsl/drivers
/dev/sdd        1007G  1.8G  954G  1% /
none            3.8G  36K  3.8G  1% /mnt/wslg
none            3.8G  0  3.8G  0% /usr/lib/wsl/lib
rootfs          3.8G  2.8M  3.8G  1% /init
none            3.8G  552K  3.8G  1% /run
none            3.8G  0  3.8G  0% /run/lock
none            3.8G  0  3.8G  0% /run/shm
none            3.8G  80K  3.8G  1% /mnt/wslg/versions.txt
none            3.8G  80K  3.8G  1% /mnt/wslg/doc
C:\             129G  11G   19G  87% /mnt/c
D:\             348G  315M  347G  1% /mnt/d
none            1.0M  0  1.0M  0% /run/credentials/systemd-journald.service
tmpfs           3.8G  0  3.8G  0% /tmp
none            1.0M  0  1.0M  0% /run/credentials/systemd-resolved.service
none            1.0M  0  1.0M  0% /run/credentials/getty@tty1.service
tmpfs           766M  12K  766M  1% /run/user/1000
jella_pragathi@LAPTOP-RQQT555:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           7.5Gi        573Mi        6.4Gi         3.5Mi        685Mi        6.9Gi
Swap:          2.0Gi           0B          2.0Gi
```

```
jella_pragathi@LAPTOP-RQQT555:~$ ps -ef
UID          PID    PPID  C  TIME TTY          TIME CMD
root          1        0  0  08:13 ?        00:00:00 /sbin/init
root          2        1  0  08:13 hvc0    00:00:00 /init
root          7        2  0  08:13 hvc0    00:00:00 plan9 --control-socket 7
root          47       1  0  08:13 ?        00:00:00 /usr/lib/systemd/systemd
systemd+      82       1  0  08:13 ?        00:00:00 /usr/lib/systemd/systemd
root          90       1  0  08:13 ?        00:00:00 /usr/lib/systemd/systemd
root         111       1  0  08:13 ?        00:00:00 /bin/sh /usr/lib/systemd
root         112       1  0  08:13 ?        00:00:00 /usr/sbin/cron -f -P
message+     113       1  0  08:13 ?        00:00:00 @dbus-daemon --system --
root         152       1  0  08:13 ?        00:00:00 /usr/bin/python3 /usr/bi
root         174       1  0  08:13 ?        00:00:00 /usr/lib/systemd/systemd
root         177       1  0  08:13 ?        00:00:00 /usr/libexec/wsl-pro-ser
syslog       193       1  0  08:13 ?        00:00:00 /usr/sbin/rsyslogd -n -i
root         210       1  0  08:13 tty1    00:00:00 /usr/sbin/agetty --nores
_chrony      227      111  0  08:13 ?        00:00:00 /usr/sbin/chronyd -n -F
_chrony      244      227  0  08:13 ?        00:00:00 /usr/sbin/chronyd -n -F
root         269       1  0  08:13 ?        00:00:00 /usr/bin/python3 /usr/sh
root         326       2  0  08:13 ?        00:00:00 /init
root         327      326  0  08:13 ?        00:00:00 /init
jella_p+     332      327  0  08:13 pts/0    00:00:00 -bash
root         334       2  0  08:13 ?        00:00:00 login -- jella_pragathi
jella_p+     388       1  0  08:13 ?        00:00:00 /usr/lib/systemd/systemd
jella_p+     391      388  0  08:13 ?        00:00:00 (sd-pam)
jella_p+     420      334  0  08:13 pts/1    00:00:00 -bash
jella_p+     670      332  0  08:17 pts/0    00:00:00 ps -ef
```

```
jella_pragathi@LAPTOP-RQQT555:~$ top
top - 08:20:42 up 7 min, 1 user, load average: 0.09, 0.05, 0.01
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 99.9 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7651.2 total, 6446.0 free, 608.1 used, 749.3 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7043.1 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   24476   15528 11620 S   0.0   0.2   0:00.84 systemd
    2 root        20   0   3180    2208  2072 S   0.0   0.0   0:00.01 init-systemd(Ub
    7 root        20   0   3180    2044  1940 S   0.0   0.0   0:00.00 init
   47 root        19  -1  50424   16732 15464 S   0.0   0.2   0:00.20 systemd-journal
   82 systemd+    20   0   22540   14640 12020 S   0.0   0.2   0:00.08 systemd-resolve
   90 root        20   0   35444   12396  9224 S   0.0   0.2   0:00.38 systemd-udevd
  111 root        20   0   2888    1952  1820 S   0.0   0.0   0:00.05 chronyd-starter
  112 root        20   0   4472    3168  2924 S   0.0   0.0   0:00.00 cron
  113 message+    20   0   8800    5408  4720 S   0.0   0.1   0:00.05 dbus-daemon
  152 root        20   0   44092   29704 14564 S   0.0   0.4   0:00.30 networkd-dispat
  174 root        20   0   18436   9448   8200 S   0.0   0.1   0:00.11 systemd-logind
  177 root        20   0 1792300   15380 12552 S   0.0   0.2   0:00.12 wsl-pro-service
  193 syslog      20   0  220548   5940   4624 S   0.0   0.1   0:00.09 rsyslogd
  210 root        20   0   5492    2724  2540 S   0.0   0.0   0:00.01 agetty
  227 _chrony      20   0   24252  11780  7236 S   0.0   0.2   0:00.09 chronyd
  244 _chrony      20   0   12092   2404   1760 S   0.0   0.0   0:00.00 chronyd
  269 root        20   0 123456   32632 18024 S   0.0   0.4   0:00.20 unattended-upgr
  326 root        20   0   3184    1112   980 S   0.0   0.0   0:00.00 SessionLeader
  327 root        20   0   3200    1248  1108 S   0.0   0.0   0:00.02 Relay(332)
  332 jella_p+    20   0   6268   5536   3864 S   0.0   0.1   0:00.13 bash
  334 root        20   0   8648   5604   4760 S   0.0   0.1   0:00.01 login
  388 jella_p+    20   0  22508  13336 10892 S   0.0   0.2   0:00.15 systemd
  391 jella_p+    20   0  22912   4088   2084 S   0.0   0.1   0:00.00 (sd-pam)
  420 jella_p+    20   0   6136   5400   3852 S   0.0   0.1   0:00.04 bash
  680 jella_p+    20   0   8172   6032   3920 R   0.0   0.1   0:00.11 top
```

```
jella_pragathi@LAPTOP-RQQGT555:~$ vmstat
procs -----memory----- --swap-- -----io----- -system-- -----cpu-----
 r b   swpd   free   buff  cache   si   so    bi   bo    in   cs us sy id wa st gu
  0  0       0 6598512 49852 718208    0    0     0    974    20  435    0  0  0 100  0  0  0
jella_pragathi@LAPTOP-RQQGT555:~$ iostat
Command 'iostat' not found, but can be installed with:
sudo apt install sysstat
jella_pragathi@LAPTOP-RQQGT555:~$ sudo apt update
sudo apt install sysstat -y
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:3 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:5 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [447 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [371 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [89.6 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [111 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [84.7 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [32.8 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [235 kB]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.1 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [76.9 kB]
Get:14 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [321 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [166 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [331 kB]
Get:17 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [61.2 kB]
Get:18 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [63.0 kB]
Get:19 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [11.4 kB]
Get:20 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse Translation-en [3032 B]
Fetched 2723 kB in 8s (333 kB/s)
110 packages can be upgraded. Run 'apt list --upgradable' to see them.
Installing:
  sysstat

Suggested packages:
  isag

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 110
  Download size: 517 kB
```

```
jella_pragathi@LAPTOP-RQQGT555:~$ iostat
Linux 6.18.33.2-microsoft-standard-WSL2 (LAPTOP-RQQGT555)      08/07/26   _x86_64_ (18 CPU)

avg-cpu:  %user   %nice %system %iowait  %steal   %idle
           0.09    0.00    0.12    0.07    0.00   99.72

Device            tps    kB_read/s    kB_wrtn/s    kB_dscd/s    kB_read    kB_wrtn    kB_dscd
sda                0.97       215.17         0.00         0.00     162257         0         0
sdb                0.20         9.29         0.00         0.00       7005         0         0
sdc                0.13         2.61         0.01         0.00       1968         4         0
sdd               14.46       671.93       212.21       667.14     506705     160028     503092

jella_pragathi@LAPTOP-RQQGT555:~$ lscpu
Architecture:        x86_64
CPU op-mode(s):      32-bit, 64-bit
Address sizes:        46 bits physical, 48 bits virtual
Byte Order:           Little Endian
CPU(s):              18
On-line CPU(s) list: 0-17
Vendor ID:            GenuineIntel
Model name:           Intel(R) Core(TM) Ultra 5 125H
CPU family:           6
Model:               170
Thread(s) per core:   2
Core(s) per socket:   9
Socket(s):            1
Stepping:             4
BogoMIPS:             5990.40
Flags:                fpu vme de pse tsc msr pae mce cx8 apic sep mtr
                    r pge mca cmov pat pse36 clflush mmx fxsr sse s
                    se2 ss ht syscall nx pdpe1gb rdtscp lm constant
                    _tsc rep_good nopl xtopology tsc_reliable nonst
                    op_tsc cpuid tsc_known_freq pni pclmulqdq vmx s
                    sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p
                    opcnt tsc_deadline_timer aes xsave avx f16c rdr
                    and hypervisor lahf_lm abm 3dnowprefetch ssbd i
                    brs ibpb stibp ibrs_enhanced tpr_shadow ept vpi
                    d ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi
                    2 erms invpcid rdseed adx smap clflushopt clwb
                    sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
```

```

2 erms invpcid rdseed adx smap clflushopt clwb
sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
vnm1 umip waitpkg gfni vaes vpclmulqdq rdpid mo
vdiri movdir64b fsrcm md_clear serialize ibt flu
sh_lld arch_capabilities
Virtualization features:
  Virtualization: VT-x
  Hypervisor vendor: Microsoft
  Virtualization type: full
Caches (sum of all):
  L1d: 432 KiB (9 instances)
  L1i: 576 KiB (9 instances)
  L2: 18 MiB (9 instances)
  L3: 18 MiB (1 instance)
NUMA:
  NUMA node(s): 1
  NUMA node0 CPU(s): 0-17
Vulnerabilities:
  Gather data sampling: Not affected
  Ghostwrite: Not affected
  Indirect target selection: Not affected
  Itlb multihit: Not affected
  L1tf: Not affected
  Mds: Not affected
  Meltdown: Not affected
  Mmio stale data: Not affected
  Old microcode: Not affected
  Reg file data sampling: Not affected
  Retbleed: Mitigation; Enhanced IBRS
  Spec rstack overflow: Not affected
  Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
  Spectre v2: Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-IBRS SW sequence; BHI BHI_DIS_S
  Srbds: Not affected
  Tsa: Not affected
  Tsx async abort: Not affected
  Vmscape: Not affected

```

```

jella_pragathi@LAPTOP-RQQGT555:~$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda    8:0    0 356.9M  1 disk
sdb    8:16   0 159.4M  1 disk
sdc    8:32   0    2G    0 disk [SWAP]
sdd    8:48   0    1T    0 disk /mnt/wslg/distro
/

```

```

jella_pragathi@LAPTOP-RQQGT555:~$ mount
none on /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2 type overlay (rw,nosuid,nodev,noatime,lowerdir=/modules,upperdir=/lib/modules/6.18.33.2-microsoft-standard-WSL2/rw/upper,workdir=/lib/modules/6.18.33.2-microsoft-standard-WSL2/rw/work,uuid=on)
none on /mnt/wsl type tmpfs (rw,relatime)
drivers on /usr/lib/wsl/drivers type 9p (ro,nosuid,nodev,noatime,aname=drivers,fmask=222,dmask=222,cache=0x5,access=client,msize=65536,trans=fd,rfd=8,wfd=8)
/dev/sdd on / type ext4 (rw,relatime,discard,errors=remount-ro,data=ordered)
none on /mnt/wslg type tmpfs (rw,relatime)
/dev/sdd on /mnt/wslg/distro type ext4 (ro,relatime,discard,errors=remount-ro,data=ordered)
none on /usr/lib/wsl/lib type overlay (rw,nosuid,nodev,noatime,lowerdir=/gpu_lib_packaged:/gpu_lib_inbox,upperdir=/gpu_lib/rw/upper,workdir=/gpu_lib/rw/work,uuid=on)
rootfs on /init type rootfs (ro,size=3911632k,nr_inodes=977908)
none on /dev type devtmpfs (rw,nosuid,relatime,size=3911632k,nr_inodes=977908,mode=755)
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,noatime)
proc on /proc type proc (rw,nosuid,nodev,noexec,noatime)
devpts on /dev/pts type devpts (rw,nosuid,noexec,noatime,gid=5,mode=620,ptmxmode=000)
none on /run type tmpfs (rw,nosuid,nodev,mode=755)
none on /run/lock type tmpfs (rw,nosuid,nodev,noexec,noatime)
none on /run/shm type tmpfs (rw,nosuid,nodev,noatime)
none on /dev/shm type tmpfs (rw,nosuid,nodev,noatime)
none on /run/user type tmpfs (rw,nosuid,nodev,noexec,noatime,mode=755)
binfmt_misc on /proc/sys/fs/binfmt_misc type binfmt_misc (rw,relatime)
cgroup2 on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime,nsdelegate)
none on /mnt/wslg/versions.txt type overlay (rw,relatime,lowerdir=/systemvhd,upperdir=/system/rw/upper,workdir=/system/rw/work,uuid=on)
none on /mnt/wslg/doc type overlay (rw,relatime,lowerdir=/systemvhd,upperdir=/system/rw/upper,workdir=/system/rw/work,uuid=on)
none on /tmp/.X11-unix type tmpfs (ro,relatime)
C:\ on /mnt/c type 9p (rw,noatime,aname=drvfs,path=C:\;uid=1000,gid=1000,symlinkroot=/mnt/,cache=0x5,access=client,msize=65536,trans=fd,rfd=6,wfd=6)
D:\ on /mnt/d type 9p (rw,noatime,aname=drvfs,path=D:\;uid=1000,gid=1000,symlinkroot=/mnt/,cache=0x5,access=client,msize=65536,trans=fd,rfd=6,wfd=6)
none on /run/user type tmpfs (rw,relatime)
none on /run/credentials/systemd-journald.service type tmpfs (ro,nosuid,nodev,noexec,relatime,nosymfollow,size=1024k,nr_inodes=1024,mode=700,noswap)
mqueue on /dev/mqueue type mqueue (rw,nosuid,nodev,noexec,relatime)
tmpfs on /tmp type tmpfs (rw,nosuid,nodev,size=3917392k,nr_inodes=1048576)
tracefs on /sys/kernel/tracing type tracefs (rw,nosuid,nodev,noexec,relatime)
hugetlbfs on /dev/hugepages type hugetlbfs (rw,nosuid,nodev,relatime,pagesize=2M)
debugfs on /sys/kernel/debug type debugfs (rw,nosuid,nodev,noexec,relatime)
fusectl on /sys/fs/fuse/connections type fusectl (rw,nosuid,nodev,noexec,relatime)
configfs on /sys/kernel/config type configfs (rw,nosuid,nodev,noexec,relatime)
none on /run/credentials/systemd-resolved.service type tmpfs (ro,nosuid,nodev,noexec,relatime,nosymfollow,size=1024k,nr_inodes=1024,mode=700,noswap)
none on /tmp/.X11-unix type tmpfs (ro,relatime)
none on /run/credentials/getty@ttyL.service type tmpfs (ro,nosuid,nodev,noexec,relatime,nosymfollow,size=1024k,nr_inodes=1024,mode=700,noswap)
tmpfs on /run/user/1000 type tmpfs (rw,nosuid,nodev,relatime,size=783476k,nr_inodes=195869,mode=700,uid=1000,gid=1000)
tmpfs on /mnt/wslg/run/user/1000 type tmpfs (rw,nosuid,nodev,relatime,size=783476k,nr_inodes=195869,mode=700,uid=1000,gid=1000)

```

```
jella_pragathi@LAPTOP-RQQT555:~$ history
```

```
1  pwd
2  clear
3  pwd
4  histroy
5  date
6  clear
7  whoami
8  cal
9  pwd
10 whoami
11 hostname
12 uname-a
13 mkdir OSSP
14 cd OSSP
15 git init
16 mkdir src include obj bin docs tests scripts assets
17 touch .gitignore README.md LICENSE Makefile
18 touch src/{main.c,shell.c,parser.c,executor.c,builtin.c}
19 touch include/{shell.h,parser.h,executor.h,builtin.h}
20 touch bin/my_shell
21 touch docs/{design.md,report.pdf}
22 touch tests/{test_parser.c,test_executor.c}
23 touch scripts/run.sh
24 chmod +x scripts/run.sh
25 touch assets/architecture.png
26 tree
27 sudo apt install tree
28 mkdir OSSP
29 cd OSSP
30 git init
31 mkdir src include obj bin docs tests scripts assets
32 touch .gitignore README.md LICENSE Makefile
33 touch src/{main.c,shell.c,parser.c,executor.c,builtin.c}
34 touch include/{shell.h,parser.h,executor.h,builtin.h}
35 touch bin/my_shell
36 touch docs/{design.md,report.pdf}
37 touch tests/{test_parser.c,test_executor.c}
38 touch scripts/run.sh
39 chmod +x scripts/run.sh
40 touch assets/architecture.png
```

```
62  ps
63  ps -ef
64  top
65  pwd
66  history
67  pwd
68  ls
69  date
70  whoami
71  hostname
72  uname
73  uname -r
74  id
75  tty
76  uptime
77  users
78  who
79  w
80  groups
81  env
82  printenv
83  df -h
84  free -h
85  ps -ef
86  top
87  q
88  vmstat
89  iostat
90  sudo apt update
91  sudo apt install sysstat -y
92  iostat
93  lscpu
94  lsblk
95  mount
96  history
jella_pragathi@LAPTOP-RQQT555:~$ |
```

```
jella_pragathi@LAPTOP-RQQT555:~$ echo "Hello World"
Hello World
jella_pragathi@LAPTOP-RQQT555:~$ sudo apt install bc -y
[sudo: authenticate] Password:
bc is already the newest version (1.07.1-4build1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 110
jella_pragathi@LAPTOP-RQQT555:~$ bc
bc 1.07.1
Copyright 1991-1994, 1997, 1998, 2000, 2004, 2006, 2008, 2012-2017 Free Software Foundation, Inc.
This is free software with ABSOLUTELY NO WARRANTY.
For details type 'warranty'.
^C
(interrupt) use quit to exit.
5+10
20*3
100/4
quit
15
60
25
jella_pragathi@LAPTOP-RQQT555:~$ factor 100
100: 2 2 5 5
jella_pragathi@LAPTOP-RQQT555:~$ sleep 5
jella_pragathi@LAPTOP-RQQT555:~$
```